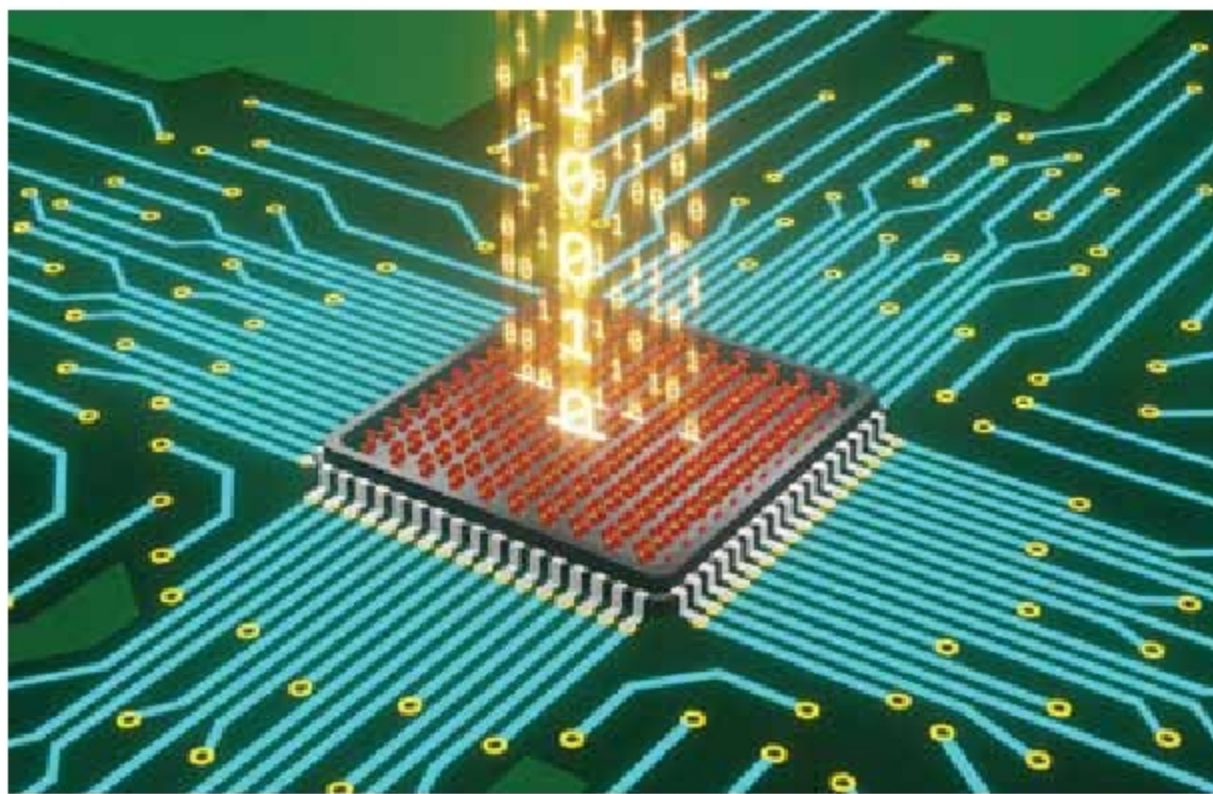




Smarter artificial intelligence technology in a new light-powered chip



A graphic illustration showing how the technology combines the core software needed to drive AI with image-capturing hardware, in a single electronic device (Credit: RMIT University)

Researchers at the RMIT University have developed artificial intelligence technology that brings together imaging, processing, machine learning, and memory in one electronic chip, powered by light. The prototype shrinks artificial intelligence technology by imitating the way the human brain processes visual information.

The nanoscale advance combines the core software needed to drive artificial intelligence with image-capturing hardware in a single electronic device. With further development, the light-driven prototype could enable smarter and smaller autonomous technologies like drones and robotics, plus smart wearables and bionic implants like artificial retinas.

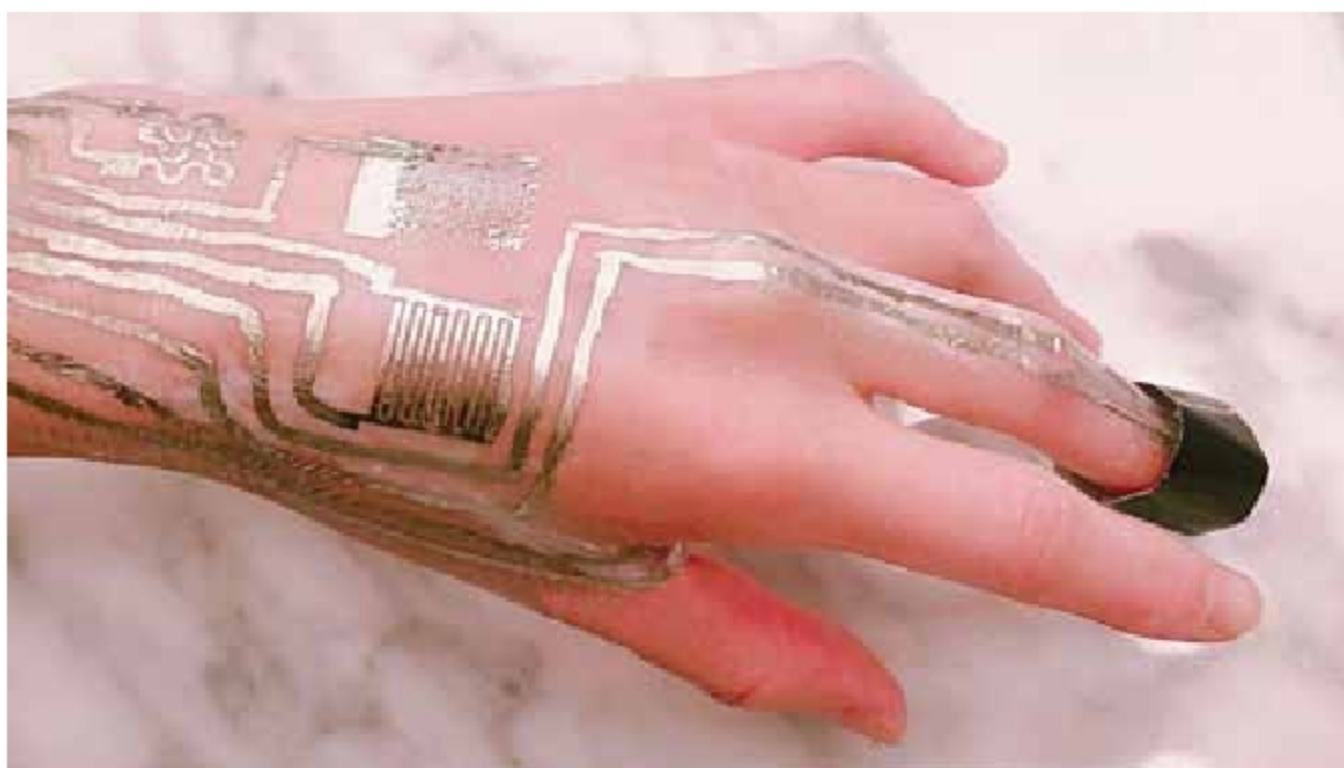
The device is readily compatible with existing electronics and silicon technologies, for effortless future integration.

Wearable, high-performance biometric sensors can now be printed directly on skin

An international team of researchers has developed a novel technique to produce precise, high-performing biometric sensors.

Flexible printed circuit boards for use in wearable sensors had been previously developed, but printing directly on skin has been hindered by the bonding process for the metallic components in the sensor. Called sintering, this process typically requires temperatures of around 300 degrees Celsius to bond the sensor's silver nanoparticles together.

Since the skin surface cannot withstand such a high temperature, a room temperature sintering aid layer consisting of polyvinyl alcohol paste was used. The layer reduced printing surface roughness and allowed for an ultrathin layer of metal patterns that can be bent and folded while maintaining electromechanical capabilities.



With a novel layer to help the metallic components of the sensor bond, an international team of researchers printed sensors directly on human skin (Credit: Ling Zhang, Penn State/Cheng Lab and Harbin Institute of Technology)